Step 2a - H3 - Microwave user efficiency simulation

Market scan

Market size

The underlying installed-equipment market is small but real: global industrial microwave heating equipment sales are ~$1.4-1.6B/yr (2025, ~4-6% CAGR), which implies an installed base of only tens of thousands of machines. The directly addressable software/optimization layer for that installed base is a niche — likely ~$50-200M/yr at best today (~1-10% of equipment spend on simulation/optimization services). The adjacent, much larger context is the process digital-twin market (~$2.6B in 2025, ~31% CAGR), which shows strong operator demand for efficiency software but is dominated by generic platforms without microwave physics.

Region	Size ($B)	Region	Size ($B)
Europe	~$0.35-0.45B (equipment market proxy; ~20-30% share, strong density of microwave OEMs and food/pharma users)	Asia	~$0.5-0.55B (~35% share, largest region per Grand View Research)
America	~$0.4-0.45B (~25-30% share)	Africa	~$0.1-0.15B (incl. Middle East, ~5-10% share)

Target customers

Customers are the industrial plants operating installed microwave systems (food tempering/drying/sterilizing, pharma, chemicals, wood, ceramics, rubber vulcanization). SAIREM alone claims ~5,000 machines installed in 70+ countries, so the worldwide installed base is plausibly ~10^4-10^5 machines spread over ~10,000-40,000 plants. Most operators are SMEs with no microwave engineering staff; they are hard to reach directly and are best approached through equipment OEMs, industry associations (AMPERE, IMPI), and retrofit/maintenance channels.

Customer size	Approx. number	Market share (in %)
Large	~1,000-3,000 plants	~45-55% of installed microwave capacity (multi-line multinational food/pharma/chemical operators)
Medium	~5,000-15,000 plants	~30-40%
Small	~10,000-25,000 plants	~10-20%

Players offering a solution today

No vendor today sells 'efficiency simulation for existing industrial microwave machines' as a product. The de-facto incumbents are (a) the equipment OEMs' own after-sales/service engineering, (b) generalist EM/multiphysics tools (Ansys, COMSOL, CST) usable only by specialists, and (c) generic digital-twin/APC platforms (AVEVA, AspenTech, Siemens) that lack EM-thermal microwave physics. Niche microwave-heating simulation exists only in QWED's QuickWave heating module and in consultancies/academic groups (e.g., AMPERE network). Competition is therefore fragmented and mostly indirect — the market is essentially unserved as a product category.

Player	Type of solution	Market share (in %)

Large (Ansys/Synopsys, COMSOL, Dassault SIMULIA CST, Siemens, AVEVA/AspenTech)	general EM/multiphysics suites and generic process digital-twin/APC platforms; not microwave-operation specific	~70-80% of EM-simulation spend sits with generalist vendors, but ~0% purpose-built for MW-machine operators
Medium (QWED/QuickWave, solid-state retrofit vendors e.g. RFHIC/LEANFA/SAIREM/Muegge service, specialist MW consultancies)	microwave-heating-capable FDTD simulation, closed-loop/solid-state upgrades, engineering services	~10-20% niche
Small (independent microwave consultants, university spin-offs, open-source openEMS/MEEP users)	one-off simulation studies, dielectric measurement services, custom in-house models	~5-10%

Regulation

Regulation	Jurisdiction	What it means for us
IEC/EN 60519-6	International/EU	Safety standard for industrial electroheating installations using HF/microwave energy (3 MHz-300 GHz): any retrofit or operating-point change must keep leakage and interlock compliance — a simulation product can support re-verification
CISPR 11 / EN 55011	International/EU (EMC)	Limits RF disturbance emissions from ISM equipment; changing operating frequency/power or adding solid-state sources requires re-assessment of emissions — simulation of leakage/harmonics is directly relevant
ITU Radio Regulations ISM bands	Global, region-dependent	Microwave process equipment effectively restricted to ISM frequencies (e.g., 915 MHz mainly Region 2, 2.45 GHz worldwide); regional frequency differences constrain which optimization levers (frequency tuning) are legal per market
Directive 2013/35/EU (EMF worker exposure)	EU	Sets minimum health/safety requirements for worker exposure to electromagnetic fields; stronger field coupling near machines must be checked when processes are re-tuned

Machinery Directive 2006/42/EC	EU	Substantial modification of installed microwave machines (retrofit generators, applicator changes) can trigger re-certification obligations

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
Name	Devin (agent draft)	Devin (agent draft)	Date	2026-10-30	2026-10-30	Name			Date	
Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis